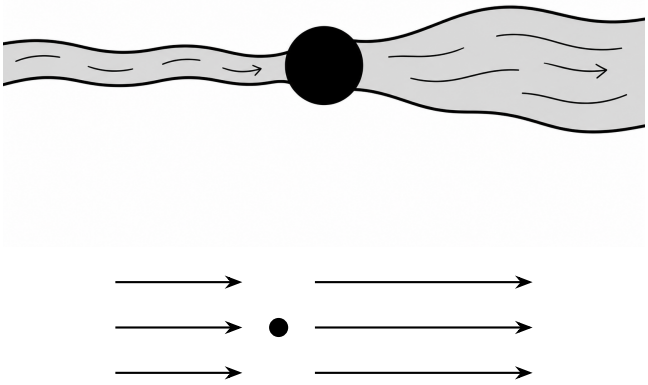
Divergence

There are two notions of the 'derivative' of a vector field, the first being the divergence. Divergence is calculated at each point in a vector field. The divergence at a point can be thought of as a small water hole around that point, where water can flow in and out.



In this water hole analogy, the direction of the vector field represents the direction of flowing water, and the magnitude represents the intensity. The graphic shows a thin stream of water entering the point and a wider stream of water exiting. This is representative of a positive divergence. Using the corresponding vector analogy, a vector field points toward a point and increases in magnitude on the other side of the point, creating a net outwards flow. The vector field at this point has a positive divergence

The divergence at a particular point is the net water flow. The divergence is positive if the net flow results in the hole acting as a water source, where the net flow is out of the hole. The divergence becomes increasingly positive the higher the intensity of the net outwards water flow. Vice versa, a negative divergence means the net water flow acts as a water sink, where the net water flow is into the hole. The magnitude becomes increasingly negative the higher the intensity of the net inwards water flow. The divergence will be zero if the the total outwards water flow exactly offsets the inwards flow, resulting in a net zero flow.

This concept has been presented for a two dimensional vector field, but the concept extends analogously to three dimensions. For a three-dimensional vector field $\mathbf{F}(x, y, z) = (f_x, f_y, f_z)$, the formula for divergence is

$$\mathbf{div}(\mathbf{F}(x, y, z)) = \frac{\partial f_x}{\partial x} + \frac{\partial f_y}{\partial y} + \frac{\partial f_z}{\partial z}$$

The alternate form involves the use of the grad operator ∇. We have seen ∇ used for $\nabla f(\mathbf{x})$ to calculate the direction of steepest ascent at the point $\mathbf{x}$. We have also seen ∇ used for ∇f, to give the vector field where each vector at each point corresponds to the direction of steepest ascent. In addition to this, mathematicians will also use the symbol ∇ on its own as the vector

$$\begin{pmatrix} \dfrac{\partial}{\partial x} \\ \dfrac{\partial}{\partial y} \\ \dfrac{\partial}{\partial z} \end{pmatrix}$$

which is neither a vector or a vector field. However, ∇ on its own can be used in conjunction with the vector field

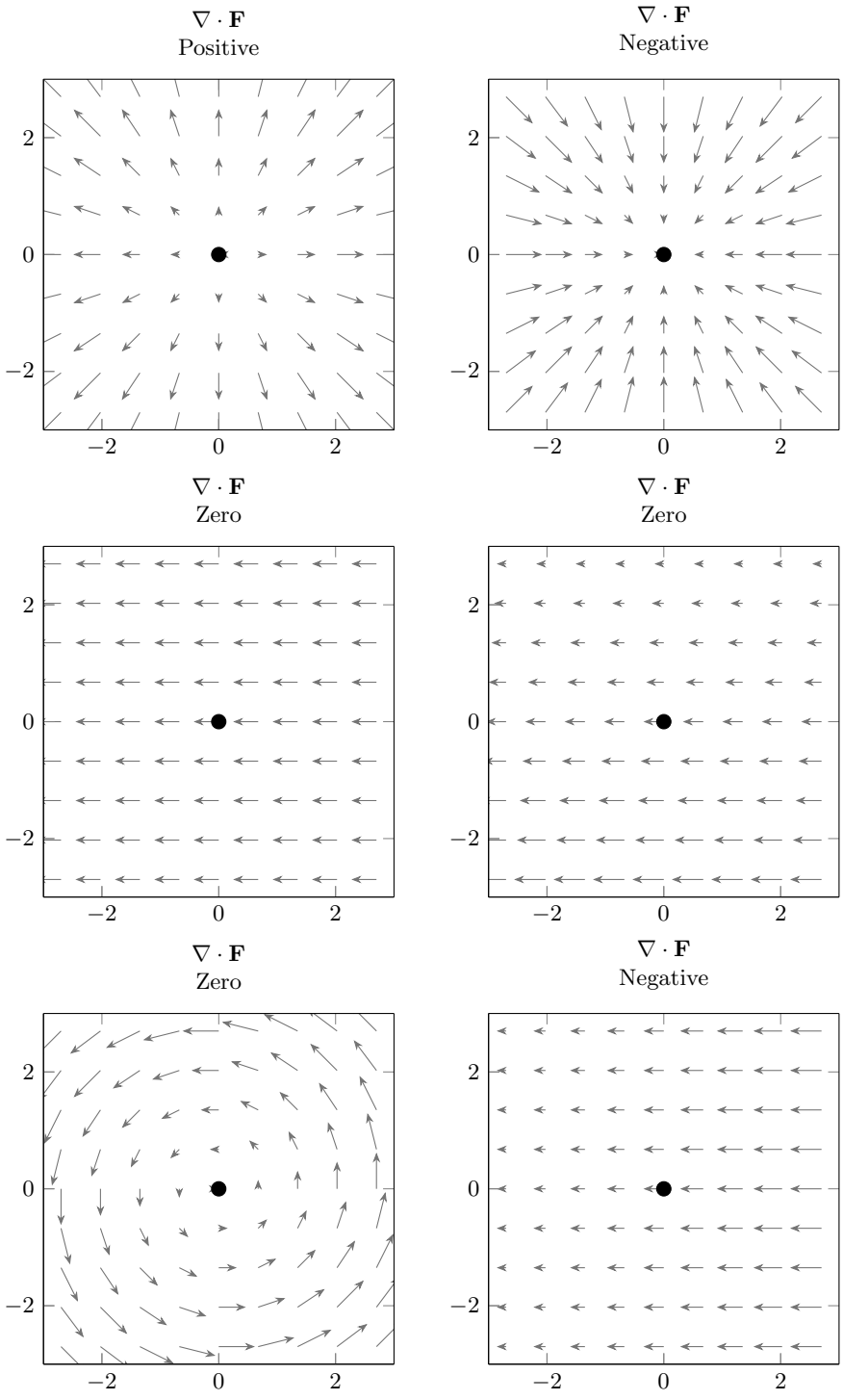
$$\mathbf{F} = \begin{pmatrix} f_x \\ f_y \\ f_z \end{pmatrix}$$

and together with a dot product they produce

$$\nabla \cdot \mathbf{F} = \begin{pmatrix} \dfrac{\partial}{\partial x} \\ \dfrac{\partial}{\partial y} \\ \dfrac{\partial}{\partial z} \end{pmatrix} \cdot \begin{pmatrix} f_x \\ f_y \\ f_z \end{pmatrix} = \frac{\partial f_x}{\partial x} + \frac{\partial f_y}{\partial y} + \frac{\partial f_z}{\partial z}$$

This convention is often why the formula for divergence is written as

$$\mathbf{div}(\mathbf{F}) = \nabla \cdot \mathbf{F}$$


$\nabla \cdot \mathbf{F}$
Positive
$\nabla \cdot \mathbf{F}$
Negative
$\nabla \cdot \mathbf{F}$
Zero
$\nabla \cdot \mathbf{F}$
Zero
$\nabla \cdot \mathbf{F}$
Zero
$\nabla \cdot \mathbf{F}$
Negative
2
0
−2
−2
0
2

Calculating the divergence at each point will output a regular scalar function $f : \mathbb{R}^2 \to \mathbb{R}$, which you can visualise using a heat map, or in the case of two dimensional vector fields, a surface in 3D.

